Course code: **V23CST15** **V23** HTNO | | | | | | | | | | |

SRI VASAVI ENGINEERING COLLEGE (Autonomous)

B. Tech VII Semester Regular Examinations, Nov/Dec-2026

(Model Paper-I)

DEEP LEARNING

(Computer Science & Engineering)

Time: 3 Hrs Max. Marks: 70

			PART-A **Answer All the Questions.**	
1.				**20 M**
	i.		What is Data Preprocessing?	CO1-K1(2M)
	ii.		Define Overfitting.	CO1-K1(2M)
	iii.		What is Backpropagation?	CO2-K1(2M)
	iv.		Name any four optimization algorithms used in Deep Learning.	CO2-K1(2M)
	v.		What is Binary Classification?	CO3-K1(2M)
	vi.		What is Model Evaluation?	CO3-K1(2M)
	vii.		Define Pooling in CNN.	CO4-K1(2M)
	viii.		What is Padding?	CO4-K1(2M)
	ix.		What is an Autoencoder?	CO5-K1(2M)
	x.		Expand GAN and LSTM.	CO5-K1(2M)
			PART-B **All Questions Carry Equal Marks**	
2.				**10 M**
	A.	i.	Explain the fundamentals of Machine Learning.	CO1- K2(5M)
		ii.	Discuss the four branches of Machine Learning and compare ML with Deep Learning.	CO1- K2(5M)
			OR	
	B.	i.	Explain data preprocessing techniques.	CO1- K2(5M)
		ii.	Describe the importance of performance evaluation metrics with examples.	CO1- K2(5M)
3.				**10 M**
	A.	i.	Explain the structure of biological neurons.	CO2- K2(5M)
		ii.	Compare structure of biological neurons with artificial neurons.	CO2- K2(5M)
			OR	
	B.	i.	Explain cost functions, optimizers, hidden-layer tuning and model generalization in Deep Neural Networks.	CO2- K2(5M)
		ii.	Discuss the role of activation functions.	CO2- K2(5M)
4.				**10 M**
	A.	i.	Develop a Keras model for movie review binary classification and explain the implementation steps.	CO3- K3(5M)
		ii.	Develop an ANN model using Keras for binary classification.	CO3- K3(5M)
			OR	
	B.	i.	Develop a neural network model for multiclass news classification using Keras.	CO3- K3(5M)
		ii.	Explain preprocessing, training and evaluation.	CO3- K3(5M)
5.				**10 M**
		i.	Develop a CNN using PyTorch for image recognition.	CO4- K3(5M)
		ii.	Explain convolution, pooling, flattening and output layers.	CO4- K3(5M)
			OR	
	B.	i.	Develop an RNN using PyTorch for sequential data analysis	CO4- K3(5M)
		ii.	. Explain the vanishing gradient problem and its effect on learning.	CO4- K3(5M)
6.				**10 M**
	A.	i.	Explain the role of Deep Learning in Healthcare, Agriculture and NLP.	CO5- K2(10M)
			OR	
	B.	i.	Explain the concepts of Autoencoders, LSTMs and Deep Reinforcement Learning	CO5- K2(10M)
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